CELEMICS

Kit User Manual

Target Enrichment

Enhanced Hybridization Kit

—

Illumina Platform

Research Use Only

CELEMICS

Materials

Target Capture Solution #1 **Store at -20°C**

Box No.	Product Name	1 rxn (µL)	Cat No. 2 Hyb-rxns	Cat No. 4 Hyb-rxns	Cat No. 16 Hyb-rxns	Cat No. 48 Hyb-rxns
TI11	Blocking Reagent (HD)	5.6	TI11S-S-F00-02	TI11S-S-F00-04	TI11S-S-F00-16	TI11S-S-F00-48
	Target Capture Probe #2	1				
	Post Capture PCR Forward Primer	2.5				
	Post Capture PCR Reverse Primer	2.5				

Target Capture Solution #2 **Store at RT**

Box No.	Product Name	1 rxn (µL)	Cat No. 2 Hyb-rxns	Cat No. 4 Hyb-rxns	Cat No. 16 Hyb-rxns	Cat No. 48 Hyb-rxns
TI20	Wash Buffer #1	800	TI20S-E-F00-02	TI20S-E-F00-04	TI20S-E-F00-16	TI20S-E-F00-48
	Wash Buffer #2	500				
	Wash Buffer #3	1500				
	Hyb Buffer	16				
	Hyb Buffer Enhancer	2				

Target Capture Solution #3 **Store at -80°C**

Box No.	Product Name	1 rxn (µL)	Cat No. 2 Hyb-rxns	Cat No. 4 Hyb-rxns	Cat No. 16 Hyb-rxns	Cat No. 48 Hyb-rxns
TI30	Target Capture Probe #1	6	TI30-H-800-02 TI30-L-400-02	TI30-H-800-04 TI30-L-400-04	TI30-H-800-16 TI30-L-400-16	TI30-H-800-48 TI30-L-400-48

The total reagent volume depends on the number of hyb-reactions in the purchased kit. The number of hyb-reactions is determined by dividing the product unit (sample basis) by the multiplex level.

A: Product unit (sample basis)
B: Multiplex level

The number of the Hyb-rxn per kit = A/B

Additional Accessory Materials

Additionally Provided Materials List

Product Name	Cat No.	Store at (°C)
CeleMag Clean-up Bead	CMCB-M-050 (5 mL) CMCB-M-500 (50 mL)	4°C
CeleMag Streptavidin Bead	CMSB-M-200 (2 mL) CMSB-M-00A (10 mL)	4°C
CLM Polymerase	CMPF-M-500 (5 mL)	-20°C

* The All-in-One Package includes additional accessory materials with dedicated capacity according to the package option, so it may differ from the Cat No. provided in the table above.

Materials Required (Not Provided)

Not Provided Materials List

Reagents & Consumables		
Product Name	Suggested Supplier	p/n
Nuclease-free water (not DEPC-treated)	General lab supplier	-
Ethanol (200 proof)	General lab supplier	-
Conical centrifuge tubes (15 mL or 50 mL)	General lab supplier	-
P10, P20, P200, and P1000 pipettes	General lab supplier	-
Multichannel pipette	General lab supplier	-
Sterile, nuclease-free aerosol barrier pipette tips	General lab supplier	-
0.2 mL PCR Tube Strips, Low Profile and Cap Strips	General lab supplier	-
Powder-free gloves	General lab supplier	-
DNA LoBind Tubes, 1.5 mL PCR clean	Eppendorf	022431021 or equivalent
Microtubes 1.5ml Clear, Sterile	AXYGEN	MCT-150-C-S or equivalent
Eppendorf™ twin.tec™ 96-well LoBind PCR Plates, Skirted	Eppendorf	0030129512 or equivalent
Microseal 'B' adhesive seals	Bio-Rad	MSB-1001 or equivalent

Equipment		
Product Name	Suggested Supplier	p/n
TapeStation Instrument	Agilent Technologies, Inc.	G2991BA or equivalent
Magnetic separator	Invitrogen	12321D or equivalent
Thermal Cycler	General lab supplier	-
Micro-centrifuge	General lab supplier	-
Vortex mixer	General lab supplier	-

Before You Begin

Before You Begin

It is recommended to read the manual carefully to understand the experimental procedure. Please follow the text **in bold** and the text highlighted in the right "NOTE" box carefully. The number in the "NOTE" box matches the procedure on the left.

Notice

Experimental results may be affected by DNA/RNA quality. We recommend using high-quality DNA/RNA samples. Low-quality samples (e.g., FFPE tissue extracted gDNA) show variable efficiency due to chemical damage and fragmentation, and its small amounts. Therefore, depending on the sample quality and input amount, the protocol may require additional experimental optimization.

Directions for the Use of This Product

1. This product should be used for research use only.
2. Reagents and specimens should be handled cautiously to avoid potential injury and infections.
    1) Handle reagents cautiously and avoid contact with any part of the body, especially skin, eyes, or mucous membranes. In the event of reagent contact, rinse immediately with water and follow duly accredited treatment.
    2) All blood serum materials from human sources should be handled carefully and treated as potential infectious sources.
3. Genomic DNA/RNA extracted from bio-specimens such as blood, saliva, FFPE, and fresh tissue should be used as sample specimens.
4. To prevent cross-contamination, change the pipette tip between each sample or reagent.
5. Usage of reagents should be followed by the instruction. We would recommend not using reagents past their expiration dates.
6. Reagents should be stored at their suggested conditions.
    Room temperature (RT, 15-25°C), Freezer(-20°C), Deep freezer (-80°C)
7. The viscosity of the reagent is different. Slow pipetting may help with accuracy when handling viscous reagents.
8. For disposal of used or expired reagents, please contact an authorized waste disposal company in accordance with relevant laws and regulations.
9. Be cautious with specific allergenic reagents.

Please maintain a nuclease-free environment during the experiment. Wear a lab coat, disposable gloves, and protective goggles. If you have any questions or concerns, please email us at manual@celemics.com

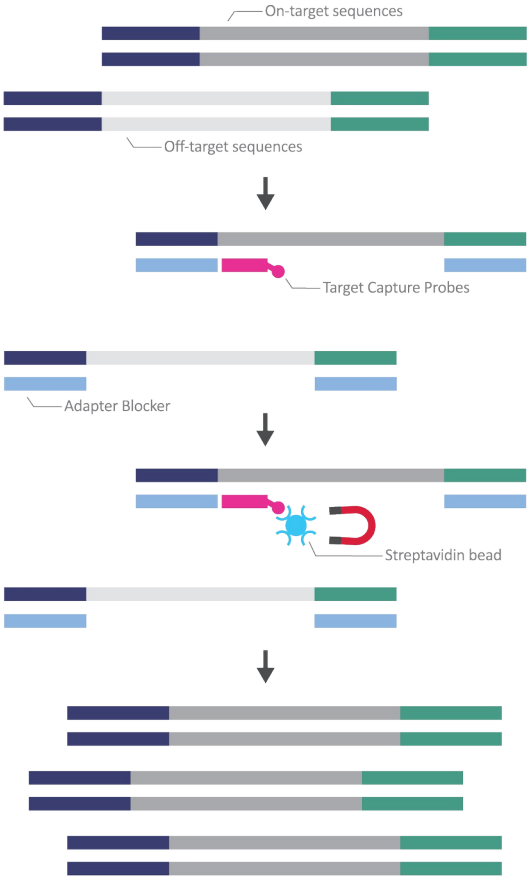
Protocol Overview

Prepare libraries and reagents for hybridization
1 hr

Hybridize target capture probes with DNA library
3 hr to 16 hr

Selection of target captured library by using streptavidin beads
1.5 hr

Amplification of the captured library
45 min



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Hybridization and Washing

Step 1. Pre-capture Multiplexing of Samples *

[Preparation]

- **CeleMag Clean-up Bead -** Place the bead at room temperature for at least 30 minutes before use. Store the bead at 4°C after use.
- **80% ethanol solution -** Using freshly made 80% ethanol solution is recommended.

Volumes of CeleMag Clean-up Bead (1X)

Reagent	Volume Ratio (Bead : Sample)
CeleMag Clean-up Bead	**1 : 1**

[Pre-capture Multiplexing Guide] **

After Assessing the quality and quantity, multiplex each sample at 500 ng per library.

Multiplexing Guide According to the Number of Samples

Number of Samples	1 DNA Library Amount (ng)	Total Sample Amount (ng)
4	500	2000
8	500	4000
12	500	6000

[Procedure]

1. Vortex CeleMag Clean-up Bead thoroughly to achieve a homogeneous state.
2. **Add each DNA library sample** to a new 1.5 mL tube according to '[Pre-capture multiplexing guide]'.
3. **Mix it with the same volume of CeleMag Clean-up Bead (1X)**. Mix thoroughly using a vortex mixer and pulse spin down the tube.
4. Incubate at room temperature for **5 minutes.**
5. Place the tube on the **magnetic separator** until the solution is clear (**3-5 minutes).**
6. Keep the tube on the magnetic separator and **discard the supernatant**.
7. While keeping the tube on the magnetic separator, **add 500 µL of fresh 80% ethanol** to each tube.
8. Incubate for 30 seconds at room temperature, then **discard ethanol**.

NOTE

* Multiplexing may be limited depending on the kit used.

** The following is an example of 4-plex.

① Calculate the volume corresponding to 500 ng and check the multiplexed sample volume.

Sample Name	Conc. (ng/µL)	Volume for 500 ng (µL)
A	100	5
B	20	25
C	25	20
D	50	10
Total volume		**60 µL**

② Use the same volume CeleMag Clean-up Bead for the multiplexed sample.

3. To avoid the bead precipitation, we do not recommend spinning down the CeleMag Clean-up bead too strongly.

6. Do not disturb the bead pellet while you remove the supernatant.

9. **Repeat wash** again (step 7-8).

10. Spin down, place the tube on the magnetic separator and **discard residual ethanol**.

11. **Air-dry the sample tube** at room temperature for **2 minutes** or until residual ethanol is completely evaporated.

12. **Add 12 µL of nuclease-free water** to the sample tube and vortex to mix. Pulse-spin down and incubate for **2 minutes** at room temperature.

13. Place the tube on the magnetic separator for **2 minutes** until the solution is clear.

14. **Transfer the supernatant (10 µL)** to a new tube.

STOP POINT:
If you are not immediately continuing to the next step, store the sample at -20°C.

NOTE

11. Overdrying of the bead pellet will decrease the elution efficiency.

Step 2. Enhanced Hybridization of the Library

IMPORTANT *

Hyb Buffer, Hyb Buffer Enhancer: Pre-warmed at 37°C before use.
Target Capture Probe #1: Thaw on ice slowly.
Do not vortex (tapping or pipetting are allowed).
Samples: Maintain at 65°C during the whole hybridization process.
Consumables
We recommend that you perform a pilot test to determine if experimental conditions are suitable for hybridization.

1. Prepare PCR plates or tubes suitable for your thermal cycler. After sealing the PCR plates or tubes with sealing tape or PCR caps, incubate 27 µL of nuclease-free water at 65°C for 2 hours (lid temperature is 105°C).

2. Ensure that there is no extensive evaporation. If the evaporation volume does not exceed 3.4 µL, the hybridization can be performed with the materials you used in the pilot test.

[Preparation]

- **Target Capture Solution Box #1** – Thaw on Ice, and keep the reagents on ice
  Blocking Reagent
  Target Capture probe #2
- **Target Capture Solution Box #2** - Pre-warmed at 37°C until the solution is clear
  Hyb Buffer
  Hyb Buffer Enhancer
- **Target Capture Solution Box #3 -** Thaw on Ice, and keep the reagents on ice
  Target Capture Probe #1 (Supplied in powder form)

[Target Capture Probe #1 Preparation Guide] *

Target Capture Probe #1 is provided in powdered form. The powdered target capture probe must be dissolved in nuclease-free water before use.

Please check the number of Hyb-rxn in your 'Target Capture Solution Box' kit and **add 8.4 µL nuclease-free water per Hyb-rxn**. Do not use a vortex mixer to dissolve the reagent. **Pipette to mix.**

Volume of nuclease-free water to be added = 8.4 * (Number of the Hyb-rxn per kit)

NOTE

* For successful results, follow the recommendations.

* **Keep** the eluted Target Capture probe #1 **on ice**.

* Once you dissolve the powder target capture probe, store the remaining probe solution at -80°C after use.

* If you need an additional help, please refer to the **'Appendix: Target Capture Probe #1 preparation guide'**

[Procedure]

Hybridization requires two reagents:

Blocking Reagent with DNA Library, **Hybridization Buffer & Probe Mix**.

These two reagents are sequentially incubated in the thermal cycler and mixed for a hybridization reaction.

1. **Preparation of Blocking Reagent with DNA Library** (pre-capture multiplexing libraries prepared using Celemics Library Prep Manual).
   a. **Transfer 10 µL of pre-capture multiplexed sample** to a new 8-strip PCR tube (or 1st column of a new 96-well PCR plate).
   b. **Add 5.6 µL of Blocking Reagent** to each tube.
   c. Mix well, seal, and spin down the tube.
   d. Keep **Blocking Reagent with DNA Library tube (15.6 µL) on ice** before hybridization.

2. **Preparation of Hybridization Buffer & Probe Mix.**

IMPORTANT:
Do not vortex Target Capture Probe #1 and Hybridization Buffer & Probe Mix

   a. Make sure that the **Hybridization Buffer and Hyb Buffer Enhancer are pre-warmed at 37°C before use.** If any precipitates remain, keep incubating at 37°C until the solution is clear.
   b. Prepare the Hybridization Buffer & Probe Mix described below. Mix well with gentle pipetting.

Hybridization Buffer & Probe Mix

Reagent	Volume for 1 Hyb Reaction (µL)
Hyb Buffer	16
Hyb Buffer Enhancer	2
Target Capture Probe #1	6
Target Capture Probe #2	1
Total Volume	**25**

   c. Transfer **25 µL Hybridization Buffer & Probe Mix** into a new 8-strip PCR tube (When using the 96-well plate, do not add the Hybridization Buffer & Probe Mix on the 96-well plate at this step).
   d. Keep **Hybridization Buffer & Probe Mix tube at RT** before hybridization.

NOTE

2-b. Prepare the Hybridization Buffer & Probe Mix right before the hybridization reaction on a thermal cycler.

2-b. When Hybridization Buffer & Probe mix are mixed, it turns into a cloudy white solution. However, this does not affect your final capture outcome.

2-d. Do not place the Hybridization Buffer & Probe mix tube on ice.

3. Perform a hybridization reaction on a thermal cycler.

   a. Set the program into a thermal cycler.

Hybridization Thermal Cycler Program (lid temperature: 105°C)

Thermal Cycler Step	Temperature	Time	Number of Cycles
1	95°C	1 minute	1
2	65°C	Hold	1
3	68°C	1 minute	120 cycles
4	60°C	3 seconds	
5	65°C	30 minutes	1
6	65°C	Up to 16 hours*	-

   b. Place the **Blocking Reagent with DNA Library tube (or plate)** on the thermal cycler and **start the Hybridization Thermal Cycler Program** (Keep the tube on the thermal cycler until 'Procedure 3-f').

   c. Once the temperature reaches 65°C (Thermal Cycler Step 2), place the **Hybridization Buffer & Probe Mix tube** into the thermal cycler and **incubate at 65°C for 1 minute**.

   d. After performing procedure 3-c (65°C for 1 minute), **open the thermal cycler** and while keeping both strip PCR tubes (or plate) at 65°C (Thermal Cycler Step 2), remove both strip PCR tube caps (or sealing film).

   e. **Transfer 15.6 µL (the entire amount) of Blocking Reagent with DNA Library into Hybridization Buffer & Probe Mix tube** (or well of 96 PCR plate). Mix well by pipetting up and down 2-3 times.

   f. **Seal the tube tightly (or PCR plate) with a new cap** (or a new sealing film) completely and **start Thermal Cycler Step 3 of the thermal cycler program.**

   g. The sample-hybridization mixture tube should be maintained at 65°C for 30 minutes during Thermal Cycler Step 5 at a minimum.

   h. Sample-hybridization mixture tube can be held at 65°C (Thermal Cycler Step 6) for up to 16 hours, until ready to proceed with 'Step 4: Selection of the Target Captured Library.'

NOTE

3-a. The **Hybridization Thermal Cycler Program** may be limited depending on the kit used.

* Thermal cycler step 6 is recommended to be incubated for a longer duration to obtain better results.

3-c. When using a 96-well PCR plate, **transfer 25 µL Hybridization Buffer & Probe Mix tube to an empty column** and quickly seal the PCR plate.

3-f. Seal the tube (or plate) tightly to avoid excessive evaporation.

3-h. After the reaction is completed, final volume of the hybridization mixture will be 30 to 32 µL. Excessive evaporation during this step can lead to Hybridization failure.

Step 3. Preparation of CeleMag Streptavidin Bead

[Preparation]

- **Target Capture Solution Box #2**
  Wash Buffer #1 - Keep at room temperature
  Wash Buffer #3 - Pre-heat at 70°C°
- **CeleMag Streptavidin Bead (CMSB) -** Store at 4°C after use

[Procedure]

1. Pre-heat Wash Buffer #3 at 70°C in a heat block for the next step (Step 4. Selection of the Target Captured Library).
2. Vortex CeleMag Streptavidin Bead to achieve a homogeneous state.
3. Add 50 µL of the CeleMag Streptavidin Bead to a new 1.5 mL LoBind tube.
4. Wash the CeleMag Streptavidin Bead as follows:
   a. **Add 200 µL of Wash Buffer #1** to the tube containing 50 µL of the CeleMag Streptavidin Bead.
   b. Mix well on the vortex mixer and spin down briefly.
   c. Place the tube in the magnetic separator until the solution is clear.
   d. Keep the tube in the magnetic separator and **discard the supernatant**.
   e. **Repeat steps a - d twice** (total 3 times).
   f. **Add 200 µL of Wash Buffer #1** to the bead.

   ⚠ **IMPORTANT:** Proceed immediately to the next step.

NOTE

4-d. Do not disturb the bead pellet while you remove the supernatant.

Step 4. Selection of the Target Captured Library

[Preparation]

- **Target Capture Solution Box #2**
  Wash Buffer #2 - Keep at room temperature
  Wash Buffer #3 - Pre-heat at 70°C

[Procedure]

1. **Keep the sample-hybridization mixture** (from 'Step 2. Enhanced Hybridization of the Library') **at 65°C,** and immediately **transfer the entire sample-hybridization mixture to the bead solution tube** (from 'Step 3. Preparation of CeleMag Streptavidin Bead').
2. Invert the tube 3-5 times to mix them thoroughly.
3. **Rotate the tube** on a rotator (25 rpm) for **30 minutes** at room temperature.
4. Spin down briefly.
5. Place the tube on a **magnetic separator until the solution is clear.**
6. Keep the tube on the magnetic separator and **discard the supernatant**.
7. Take the tube out of the magnetic separator and **add 500 µL of Wash Buffer #2** to each tube (180 µL for PCR tube or 96-well plate).
8. Mix well on a vortex mixer and pulse-spin down.
9. Incubate the tube for **15 minutes** at room temperature.
10. Briefly mix the tube on the vortex mixer and pulse spin down.
11. **Place the tube on the magnetic separator until the solution is clear** and **discard the supernatant**.
12. Rinse the bead with pre-heated Wash Buffer #3.
    a. **Add 500 µL of pre-heated Wash Buffer #3** to each tube (180 µL for PCR tube or 96-well plate).
    b. Mix well on a vortex mixer. Spin down briefly.
    c. Incubate the tube for **10 minutes at 70°C** in a heat block.
    d. Briefly mix the tube on the vortex mixer and pulse spin down.

NOTE

3. Do not vortex the tube vigorously as it may damage the Target Capture Probe.

6. Do not disturb the bead pellet while you remove the supernatant.

11. Do not disturb the bead pellet while you remove the supernatant.

e. **Place the tube on the magnetic separator until the solution is clear,** and **carefully discard the supernatant**.

f. **Repeat steps a-e twice** (total 3 times).

g. Spin down the tube for 3-5 seconds and place the tube on the magnetic separator to **discard residual Wash Buffer #3 completely**.

h. **Add 30 µL of nuclease-free water** to each tube and mix well on a vortex mixer. Spin down briefly.

i. **Now, on-bead capture pool** is prepared.

STOP POINT

If you are not immediately continuing to the next step, store the samples at -20°C.

NOTE

12-e. Do not disturb the bead pellet while you remove the supernatant.

Amplification of Target Captured Library

Step 1. Amplification of the Captured Library

[Preparation]

- **CLM Polymerase** - Place on ice (or use a cooling block)
- **Target Capture Solution Box #1** – Thaw on ice, and place on ice (or use a cooling block)
  Post Capture PCR Forward Primer
  Post Capture PCR Reverse Primer

[Procedure]

1. Prepare PCR Master Mix described below.

Volume of PCR master mix

Reagent	Volume for 1 Hyb Reaction (µL)	Volume for 8 Hyb-Reactions (µL) (with 10% overage)
CLM Polymerase	25	220
Nuclease-free water	5	44
Post Capture PCR Forward Primer	2.5	22
Post Capture PCR Reverse Primer	2.5	22
Total Volume	**35**	308

2. **Add 35 µL of the PCR Master Mix** into the new PCR tube.

3. **Add 15 µL the homogeneous on-bead capture pool** to the PCR tube.

4. Mix well and spin down briefly.

5. Place the tube (50 µL) on a thermal cycler and run the PCR program.

PCR Program

Temperature	Time	Number of Cycles
98°C	45 seconds	1
98°C	15 seconds	See **'The Number of PCR Cycles'**
60°C	15 seconds	
72°C	30 seconds	
72°C	3 minutes	1
4°C	Hold	--

NOTE

2. Make sure the mixture tube is kept on ice.

The Number of PCR Cycles

Target Size	Total Cycles
Targets Size < 1 Mb	16
Target Size ≥ 1 Mb	14

IMPORTANT: Proceed immediately to the next step.

NOTE

Step 2. Purification of Sample Using CeleMag Clean-up Bead

[Preparation]

- **CeleMag Clean-up Bead –** Place the bead at room temperature for at least 30 minutes before use. Store the bead at 4°C after use.
- **80% ethanol solution -** Using freshly made 80% ethanol solution is recommended.

Volume of CeleMag Clean-up Bead

Reagent	Volume Ratio (Bead : Sample)
CeleMag Clean-up Bead	**1.8 : 1** (e.g., 90 µL : 50 µL)

[Procedure]

1. Vortex CeleMag Clean-up Bead to achieve a homogeneous state.
2. **Add 90 µL CeleMag Clean-up Bead** to a new 1.5 mL LoBind Tube (0.2 mL PCR tube or 96-well plate is also available) and **the amplified captured DNA library (50 µL**) to the tube.
3. Mix well using a vortex mixer and pulse spin down the tube.
4. Incubate at room temperature for **5 minutes**.
5. Place the tube on the **magnetic separator until the solution is clear (3-5 minutes).**
6. Keep the tube on the magnetic separator and **discard the supernatant**.
7. While keeping the tube on the magnetic separator, **add 500 µL** of **fresh 80% ethanol** to each tube (180 µL for PCR tube or 96-well plate).
8. Incubate for 30 seconds at room temperature, then **discard ethanol**.
9. **Repeat wash** again (step 7-8).
10. Spin down, place the tube on the magnetic separator, and **discard residual ethanol**.
11. **Air-dry the sample tube** at room temperature for **2 minutes** or until residual ethanol is completely evaporated.
12. **Add 32 µL of nuclease-free water** to the sample tube and vortex to mix. Pulse spin down and **incubate for 2 minutes** at room temperature.
13. Place the tube on the magnetic separator for **2 minutes** until the solution is clear.
14. **Transfer the supernatant (30 µL)** to a new tube.

STOP POINT:
If you are not immediately continuing to the next step, store the sample at -20°C.

NOTE

3. To avoid the bead precipitation, we do not recommend spinning down the CeleMag Clean-up bead too strongly.

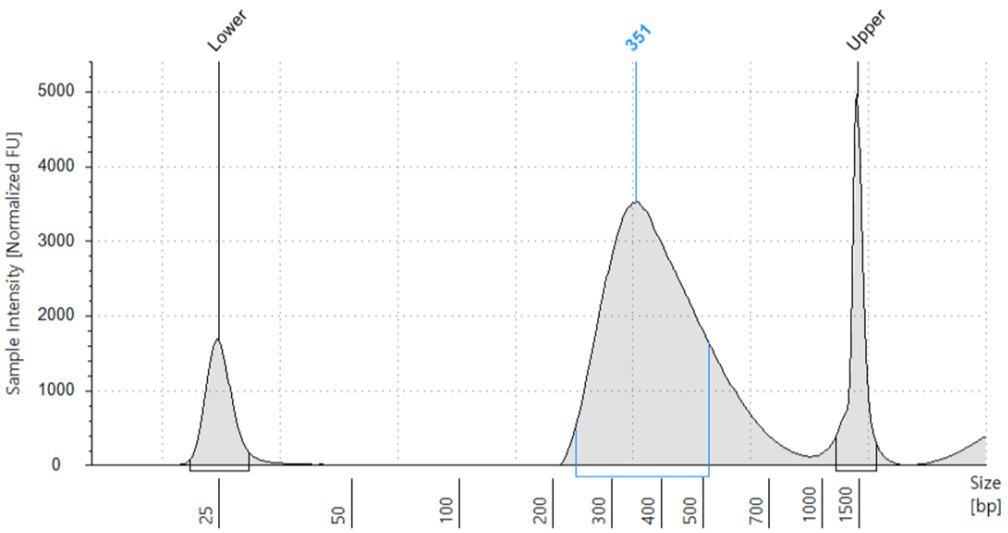
6. Do not disturb the bead pellet while you remove the supernatant.

11. Overdrying of the bead pellet will decrease the elution efficiency.

Step 3. Assess Quality and Quantity of Each Library

[Procedure]

1. Determine quality and quantity of the sample with the appropriate instrument.



This figure shows the electrophoresis diagram detected by TapeStation.

2. Make sure that the size of the amplified captured DNA library shows the main peak approximately 300-550 bp.

NOTE

Sequencing Library Pooling

In this step, all amplified captured DNA library will be pooled in a single tube. **The index of sample must not be overlapped each other.**

Consider the target size, the required data size, and the concentration of every sample.

[Procedures]

1. Create a spreadsheet as shown below.
2. Enter the values of each sample in columns 1, 2, 3, 4, 5, 6, and 8 of the spreadsheet.
3. Volume for Pooling (Column 7) is calculated using the following equation

$$Column\ 8\ (Gb) * (30\ nmol/Gb)/Column\ 5(nmol/\mu L)$$

4. Mix all amplified captured DNA library based on the calculated volumes.
5. After pooling, perform sequencing according to the running protocol of the sequencing equipment.

Example of a Spreadsheet for Sequencing Library Pooling

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8
	Contained Sample	i5 Index	i7 Index	Conc (nmol/µL)	Volume of Sample (µL)	Volume for Pooling (µL)	Require Data (Gb)
4 Plex Pre-capture Library 1	Library 1-A	501	701	91	30	6.59	20
	Library 1-B	502	702				
	Library 1-C	503	703				
	Library 1-D	504	704				
4 Plex Pre-capture Library 2	Library 2-A	501	705	45.3	30	13.25	20
	Library 2-B	502	706				
	Library 2-C	503	707				
	Library 2-D	504	708				
4 Plex Pre-capture Library 3	Library 3-A	505	701	77.2	30	7.77	20
	Library 3-B	506	702				
	Library 3-C	507	703				
	Library 3-D	508	704				
4 Plex Pre-capture Library 4	Library 4-A	505	705	61.8	30	9.71	20
	Library 4-B	506	706				
	Library 4-C	507	707				
	Library 4-D	508	708				

NOTE

3. Column 7, volume for pooling is calculated to mix 30 nmol/Gb of each amplified captured DNA library. However, those value can be changed depending on your sample's concentration.

Appendix:
Target Capture Probe #1 Preparation Guide

Target Capture Probe #1 is provided in powdered form. The powdered target capture probe must be dissolved in nuclease-free water before use.

Please check the number of your Hyb-rxn and **add 8.4 µL nuclease-free water per Hyb-rxn**. Do not use a vortex mixer to dissolve the reagent. **Pipette to mix.**

Volume of nuclease-free water to be added = 8.4 * (Number of the Hyb-rxn per kit)

Unit Size	Volume of nuclease-free water
1 Hyb-rxn	8.4 µL
2 Hyb-rxns	16.8 µL
4 Hyb-rxns	33.6 µL
8 Hyb-rxns	67.2 µL
16 Hyb-rxns	134.4 µL
32 Hyb-rxns	268.8 µL *
48 Hyb-rxns	403.2 µL *
96 Hyb-rxns	806.4 µL *

NOTE

* If the purchased kit consists of more than one tube, please divide the calculated volume by the number of tubes.